Lab Manual 1: Verification of Kirchhoff's Laws and Divider Rules in Resistive DC Circuits

1.1 Introduction and learning objectives

In this lab, students will explore the fundamental principles governing electrical circuits through hands-on experimentation with resistive DC circuits. The focus will be on verifying Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL), two cornerstone concepts in circuit analysis. Additionally, students will apply and validate the voltage divider and current divider rules, which are essential tools for analysing complex resistive networks.

Using a simple resistive circuit, students will measure voltages and currents at various points and compare the experimental results with theoretical predictions. This exercise not only reinforces analytical techniques but also builds practical skills in circuit construction, measurement, and data interpretation.

By the end of this lab, students will be able to:

1. **Apply Kirchhoff's Voltage Law (KVL)** to verify that the algebraic sum of voltages around any closed loop in a resistive circuit is zero.
2. **Apply Kirchhoff's Current Law (KCL)** to confirm that the total current entering a junction equals the total current leaving it.
3. **Use the voltage divider rule** to calculate and experimentally verify the voltage across individual resistors in a series circuit.
4. **Use the current divider rule** to determine and validate the current through individual resistors in a parallel circuit.
5. **Construct and troubleshoot basic resistive circuits** using breadboards, resistors, and DC power supplies.
6. **Accurately measure voltage and current** using a digital multimeter and compare experimental results with theoretical values.

1.2 Tools and Components Needed

Item	*Specification*
Resistors	R1 = 330 Ω, R2 = 1 kΩ, R3 = 560 Ω, R4 = 2 kΩ, R5 = 820 Ω, R6 = 1 kΩ
Voltage source	Battery or Adjustable DC Regulated Power Supply
Breadboard	Standard
Connecting Wires	Assorted
Measurement	Digital multimeter

1.3 Experimental Work

1. Always ensure the following safety checks are adhered:
   a. Power Off the supply before assembling the circuit or make changes to the circuit.
   b. Double-check connections to avoid short circuits.
   c. Use correct resistor values to prevent overheating.
   d. Avoid touching live wires while the circuit is powered.
   e. Handle multimeter probes carefully to avoid damage or incorrect readings.
2. Place resistors on the breadboard as per the circuit diagram in Figure 1.1.

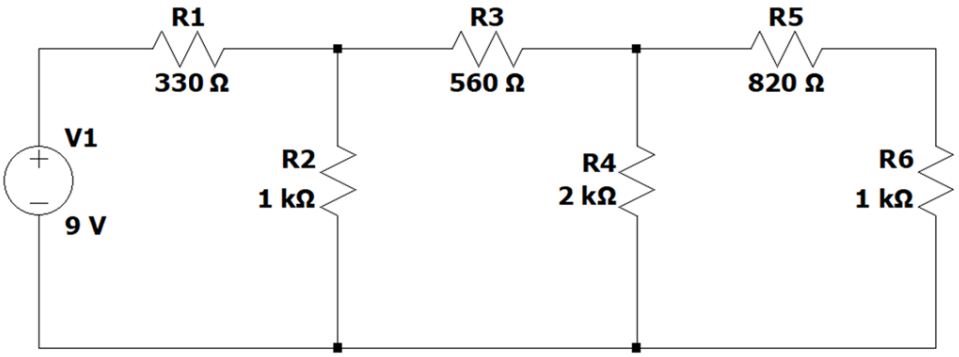


Figure 1.1

3. Connect R1 (330 Ω) in series with the positive terminal of the battery connector or voltage source **(DO NOT CONNECT THE BATTERY/TURN ON THE VOLTAGE SOURCE FIRST)**.

4. Create a common node by connecting the other end of R1 (330 Ω) to one end of R2 (1 kΩ) and R3 (560 Ω).
5. Connect the other end of R3 (560 Ω) to one end of R4 (2 kΩ) and R5 (820 Ω) to form another common node.
6. Connect the other end of R5 (820 Ω) with R6 (1 kΩ).
7. Finally, join the negative terminal of the battery connector with the remaining ends of R2, R4, and R6 as the reference node (ground).
8. Verify all connections match the circuit diagram, ask the lab executive or lecturer to check your circuit before powering on.
9. Use the multimeter for:
    a. Resistance Measurement: Set multimeter to resistor mode and place probes across the component terminals. The resistance of a resistor **MUST** be measured in isolation (not in any connection with other elements or circuit).
    b. Voltage Measurement: Set multimeter to DC Voltage mode and place probes across the component terminals. Measurement is done when the circuit is running, be careful when probing the components, short circuit may occur when probed wrongly.
    c. Current Measurement: Set multimeter to DC Current mode. Break the circuit at the point of interest. Insert multimeter in series with the component. **DO NOT** connect multimeter in parallel when measuring current.
10. Fill up the tables in the Results section.

1.4 Results and Discussion

1.4.1 Resistance Measurements

Resistor	Color code				Measured value (Ω)	% Error
	1	2	3	4		
R1 (330 Ω)						
R2 (1 kΩ)						
R3 (560 Ω)						
R4 (2 kΩ)						
R5 (820 Ω)						
R6 (1 kΩ)						

1.4.2 Voltage Measurements

Component	Measured Voltage (V)	Expected Voltage (V)	% Error
R1			
R2			
R5			
R3			
R4			
R6			

1.4.3 Current Measurements

Component	Measured Current (mA)	Expected Current (mA)	% Error
R1			
R2			
R3			
R4			
R5			
R6			

1.4.4 Calculation of expected current/voltage

1. Simplify the circuit by combining selected resistors in series/parallel, then use **voltage divider rule** to calculate the voltage across all elements, show complete working steps and equivalent circuit (hand written and drawn).
2. Simplify the circuit by combining selected resistors in series/parallel, find the total current drawn from the voltage source, then use **current divider rule** to calculate the current through all elements, show complete working steps and equivalent circuit (hand written and drawn).
3. Use **KVL** to verify the voltage across all elements and **KCL** to verify the current through all elements, show complete working steps (hand written and drawn).
4. Is there discrepancy between measured and calculated results? Discuss.

Report Marking Rubric

Criteria	Poor (1 mark)	Fair (2 marks)	Good (3 marks)	Very Good (4 marks)	Excellent (5 marks)
1. Introduction & Objectives (10%)	Vague or missing; no clear objectives	Basic overview; objectives unclear or incomplete	Adequate summary; some objectives stated	Clear introduction; most objectives covered	Comprehensive intro; all objectives clearly stated and contextualized
2. Methodology: Circuit Assembly, Safety & Measurement Technique (20%)	Circuit not assembled correctly; safety rules ignored; incorrect use of multimeter	Major errors in assembly; minimal safety adherence; some correct measurements	Circuit mostly correct; some safety steps followed; mostly correct technique	Circuit correctly assembled; most safety rules followed; reliable data with few errors	Circuit perfectly assembled; all safety protocols strictly followed; flawless technique and consistent measurements
3. Results: Data Recording & Tables (20%)	Tables missing or incomplete; data unclear	Tables present but poorly formatted or inaccurate	Tables mostly complete; some formatting or data issues	Well-organized tables; mostly accurate data	Clear, complete, and accurate tables with proper formatting
4. Results: Theoretical Calculations, Rule Application, Equivalent Circuit Construction (20%)	No calculations or major errors; no working shown; rules not applied; incorrect or missing equivalents	Attempted calculations; many errors; incomplete working; unclear rule use; unclear circuit construction	Correct approach; some errors in calculation or logic; basic rule application or equivalent circuits	Mostly accurate calculations; clear working steps; rules applied with minor errors; good circuit construction	Fully accurate calculations; all steps clearly shown and justified; rules applied accurately with clear circuit diagrams; equivalents constructed correctly
5. Discussions: Comparison, Discussion & Interpretation of Results (20%)	No comparison or discussion; lacks understanding	Minimal comparison; limited insight	Some comparison; basic discussion of discrepancies	Good comparison; thoughtful discussion of results	Thorough comparison; deep, critical analysis and excellent interpretation of discrepancies and implications
6. Conclusion, Report Structure & Presentation (10%)	No sensible conclusion. Disorganized; poor formatting; hard to follow	A conclusion is not supported by any experimental evidence. Some structure; inconsistent formatting	A conclusion is only supported by some experimental evidence. Organized; readable with minor formatting issues	A conclusion is mostly supported by experimental evidence. Well-structured; clear formatting and flow	The aim of the experiment has been achieved with evidence supported conclusion. Professionally presented; logical flow, excellent formatting and clarity

Note to students: Please attach this appendix together with the submission of coursework.